

AI-Based Short-Term Generation Forecasting of Cascaded Hydroelectric Power Plants

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Abstract—The hydraulic coupling of the Agus 1-7 hydropower cascade in Mindanao renders day-ahead capacity nomination highly sensitive to hydrologic variability and operational uncertainty. Current rule-based scheduling methods often fail to capture these non-linear interdependencies, leading to frequent re-nominations and conservative generation estimates that underutilize available water resources. To address these limitations, this study proposes an Artificial Intelligence-based short-term generation forecasting framework using a Radial Basis Function Neural Network (RBFNN). The model will integrate high-resolution historical data—including upstream releases, reservoir elevation, gate openings, and rainfall—into a sliding-window input structure to capture the dynamic behavior of the entire cascade. The methodology features a rigorous data preprocessing pipeline and chronological dataset splitting to ensure realistic performance evaluation. The study aims to benchmark the developed model against persistence and simple statistical methods using Mean Absolute Percentage Error (MAPE) to quantify improvements in prediction accuracy. Ultimately, this data-driven framework is expected to provide NPC-MinGen with a reliable tool for optimized capacity nomination, aiming to support improved grid reliability and more efficient participation in the Wholesale Electricity Spot Market.

Index Terms—Agus hydropower cascade, short-term generation forecasting, radial basis function neural network (RBFNN), cascaded hydropower plants, day-ahead forecasting, time-series modeling, capacity nomination, Mindanao power system.

I. INTRODUCTION

A. Background of the Study

Hydropower remains a cornerstone of the Philippine energy portfolio, providing a significant share of total generation and critical support for grid reliability [1]. Central to the national strategy is the Agus-Pulangi Hydroelectric Power Plant complex in Mindanao, a major government-owned facility operated by the National Power Corporation's (NPC) Mindanao Generation (MinGen) Group. The complex, with a total installed capacity of 1,001.1 MW, is sustained primarily by the hydraulically coupled Agus cascade along the Agus River, supplied by the vast storage volume of Lake Lanao [1].

The Agus cascade—comprising Agus 1, 2, 4, 5, 6, and 7 plants—is a highly interdependent system, not a collection of stand-alone units, with a combined capacity of about 746.1 MW [1]. Water released by the upstream plants immediately dictates the inflow and hydraulic head available for all downstream plants as the river flows 36 km down to Iligan Bay [1]. Consequently, any operational decision, such as changes in gate opening or unit loading at one plant, directly and non-linearly influences the water available and the potential generation capacity of every plant below it. This cascading arrangement necessitates that operation, optimization, and power generation must be planned and modeled simultaneously for the entire system, making accurate short-term forecasting essential for scheduling and market participation [1].

B. Statement of the Problem

Capacity nomination for the Agus hydropower cascade (Agus 1 to Agus 7) is frequently unstable. In practice, day-ahead nominations



Fig. 1: Schematic Representation of the Agus River Hydroelectric Power Plant (HEPP) Complex [2].

often need to be revised because the actual generation depends on a combination of hydrologic and operational factors such as forebay elevation, reservoir inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall, which are not always measured consistently or fully reflected in existing planning tools. When these inputs are incomplete, outdated, or not systematically used, the current simulation models and rule-based practices struggle to represent the true behavior of the cascade, leading to conservative nominations, frequent re-nominations, and potential underutilization of available water and capacity.

C. Objectives of the Study

The main objective is to use an AI-based short-term (one-day-ahead) generation forecasting model for the cascaded Agus 1–7 hydropower plants using an RBFNN and to investigate how these methods improve the accuracy of hydropower generation forecasting driven by data on plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall. Specifically, the study aims:

- 1) To design and develop an RBFNN-based short-term (one-day-ahead) generation forecasting model for the Agus 1–7 hydropower plants.
- 2) To evaluate the forecasting performance of the developed RBFNN model using statistical accuracy measurements using MAPE by comparing the results of the developed forecasting system with the actual data.
- 3) To compare the performance of the RBFNN model against benchmark methods to determine its improvement in prediction stability.

D. Significance of the Study

The hydraulic coupling of the Agus cascade renders day-ahead capacity nomination highly sensitive to hydrologic variability and operational uncertainty [1]. Accurate short-term generation forecasting is, therefore, an essential tool for supporting scheduling and

water management [1]. The findings of this study will benefit the following sectors:

This study will primarily benefit the National Power Corporation - Mindanao Generation (NPC-MinGen) by providing a practical, operations-oriented forecasting model that mitigates the limitations of rule-based scheduling [1]. By providing more accurate one-day-ahead forecasts, this study supports NPC-MinGen in reducing the frequency of re-nominations and minimizing the risk of unnecessary spillage. Furthermore, in the context of the Wholesale Electricity Spot Market (WESM), the proposed RBFNN framework enables data-driven decision-making for generation bidding, allowing the utility to formulate informed day-ahead offers and maximize revenue potential [1].

For the Mindanao Power Grid and System Operators, this research contributes to the overall stability and reliability of the region's energy supply [1]. By integrating upstream releases, travel time, and reservoir head dynamics into the forecast, the model provides system operators with a clearer picture of the expected hydro contribution for the next 24 hours. This improved visibility supports robust dispatch decisions, efficient reserve planning, and the maintenance of the region's energy balance [1].

The academic community and future researchers will benefit from the advancement of theoretical and practical knowledge regarding Artificial Intelligence applications in renewable energy. This study addresses the specific gap in dedicated data-driven models for coupled generation forecasting in cascaded systems [3], [4]. The methodology—specifically the rigorous preprocessing pipeline and the integration of multi-plant hydraulic features into an RBFNN architecture—serves as a reference for future researchers developing hybrid models for complex, non-linear hydrological systems [1].

Finally, this work aligns with global policy and sustainability goals, specifically the United Nations Sustainable Development Goal (SDG) 7: Affordable and Clean Energy [5]. By optimizing the operation of the Agus cascade through better forecasting, this study promotes the efficient utilization of existing renewable resources [6], thereby supporting the sustainable energy security of the region without the immediate need for new infrastructure construction [7].

E. Scope and Limitations

This study will have the following scope and limitations:

- The system under study is limited to the Agus River hydropower cascade, specifically Agus 1 to Agus 7 as operated by NPC-MinGen. Other plants in the Mindanao grid are not modeled.
- The forecasting task is restricted to short-term, one-day-ahead (24-hour) generation forecasting, using historical records of plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall. No new sensors or field measurements are installed.
- The modeling approach uses an RBFNN as the main forecasting method. Other methods, if used, serve only as simple benchmarks, and no full physical or hydrodynamic model of the Agus-Lake Lanao system is developed.
- The evaluation and implementation are limited to offline tests using historical data and statistical accuracy measures (MAPE). Real-time integration, detailed economic/market analysis, and WESM bidding simulations are outside the scope of this thesis.

F. Theoretical Framework

The present methods used in hydropower generation forecasting each have their own strengths and weaknesses. Some methods are

simple and transparent but cannot fully capture the nonlinear and coupled behavior of cascaded plants. Other methods are more flexible but difficult to tune and explain [3], [8]. This paper focuses on using an AI-based approach that can learn the relationship between hydrologic-operational variables and plant generation directly from data and give a next-day forecast with improved accuracy [9]–[12].

This is based on the concept of data-driven learning. Instead of writing down all physical equations and assumptions, the model observes many examples of past conditions—such as forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, rainfall, and plant generation—and gradually learns a mapping from these inputs to the next-day generation [3], [13], [14]. The primary objective is to produce a forecasting model that represents the actual behavior of the Agus 1–7 cascade as closely as possible and is practical for operations use [6], [15].

In this paper, the idea of data-driven learning is implemented using an RBFNN. The “participating entities” in the model are the different input variables from the Agus plants and watershed. These variables play the part of signals or predictors that the RBFNN uses to build an internal representation of the system. After training, the network becomes a forecasting tool that can accept a new set of input conditions and produce a one-day-ahead generation forecast for each plant. The theoretical framework is therefore built around four main concepts: cascaded hydropower generation, forecasting, artificial neural networks, and the proposed RBFNN-based forecasting system [16]–[19].

1) *Cascaded Hydropower Generation*: Cascaded hydropower plants share the same river flow. Water released by an upstream plant becomes the inflow of the next downstream plant. For Agus 1–7, water comes from Lake Lanao, passes through each plant in sequence, and finally discharges to Iligan Bay. The generation of each plant depends on how much water passes through its turbines and the effective head between forebay and tailwater [3].

Because of this arrangement, the plants cannot be treated as independent units. A change in forebay elevation, gate opening, or spillage at one plant can change the inflow and head of the next plant. In addition, rainfall, riparian releases, and natural inflows affect lake level and river discharge. The theoretical view of this study is that the Agus cascade behaves as one dynamic, interconnected system, where generation at any time is the result of combined hydrologic conditions and operating decisions along the river [15], [20], [21].

2) *Forecasting*: Forecasting is the process of making statements about future events whose actual outcomes have not yet been observed. In this study, the event of interest is the next-day hydropower generation of Agus 1–7. The forecast is expressed as a time series, such as hourly generation values for the next 24 hours [22], [23].

Forecasting always involves uncertainty. The goal is not to predict the future perfectly but to provide an estimate that is as close as possible to what will actually happen, given the available data. For hydropower generation, this requires using up-to-date records of plant operation and hydrologic conditions. The more accurately the model can relate past and present variables to future generations, the more useful it becomes for day-ahead capacity nomination and scheduling [24]–[27].

3) *Artificial Neural Networks and RBFNN*: Artificial Neural Networks (ANNs) are computational models inspired by the way biological neurons process information. They consist of layers of interconnected nodes that transform input data into output predictions. When used for forecasting, an ANN is trained by repeatedly

adjusting its internal parameters so that its outputs match the historical data as closely as possible [28]–[30].

An RBFNN is a particular type of ANN. It has three layers: an input layer, a hidden layer with radial basis functions (usually Gaussian), and an output layer. Each hidden neuron is “sensitive” to a certain region of the input space; it responds strongly when the current inputs are close to its center and weakly when they are far away. By combining the responses of many such neurons, an RBFNN can approximate complex, nonlinear relationships [16], [17].

In this thesis, the RBFNN is chosen because it is well suited for problems where the relationship between inputs and outputs is nonlinear but the forecasting horizon is relatively short. It can learn patterns in how generation responds to changes in forebay elevation, flows, gate positions, riparian releases, and rainfall without requiring an explicit physical model of all processes [3], [18], [19].

4) *The RBFNN-Based Forecasting System:* Figure 1 (Architecture of the RBF neural network) illustrates the theoretical operation of the proposed system. Historical data from the Agus cascade—plant generation, forebay elevation, inflow and outflow, spillway and governor gate positions, riparian flow, and rainfall—are first collected and organized. These time-series data are then pre-processed to handle missing values, remove obvious errors, and scale the variables into suitable ranges for neural-network training [31]–[34].

After preprocessing, selected past values (lags) for these variables are fed into the RBFNN as input patterns. During training, the network adjusts the centers and weights of its radial basis functions so that the predicted generation for the next day matches the historical generation as closely as possible. Once training is complete, the model can accept a new set of recent measurements and produce a one-day-ahead generation forecast for each of the Agus plants.

The resulting system can then be evaluated using statistical accuracy measures such as MAPE [35]–[37]. If the accuracy is acceptable, the forecasts may be used as support information for capacity nomination and operational planning.

5) *Model Behaviour:* The theoretical expectation is that the RBFNN will learn to mimic the typical behavior of the Agus cascade under different hydrologic and operating conditions. For example, when forebay elevation is high and inflows are strong, the model should predict higher potential generation, provided that gate positions indicate more water passing through the turbines. Conversely, when lake level is low or spillway flow is increased, the model should show a corresponding reduction in expected generation [3], [13], [14], [38].

Because the RBFNN uses many radial basis functions, it can represent different operating regimes: wet-season high inflow, dry-season low inflow, normal operation, or periods with unusual gate settings. In regions of the input space where many historical examples exist, the model is expected to give very accurate forecasts. In regions with fewer examples, the predictions may be less certain, which motivates continuous updating as new data become available [6], [24].

With this framework, the behavior of the model is intended to follow the actual behavior of the Agus cascade as closely as possible. When evaluated against historical data, the forecast curve of the trained RBFNN should closely track the observed generation curve. This match between forecast and actual behaviour is the main theoretical basis for using the RBFNN as a tool for short-term generation forecasting in the Agus 1–7 hydropower plants [35], [39].

II. REVIEW OF RELATED LITERATURE

A. Short-Term Generation Forecasting in Cascaded Hydropower Systems

The liberalization of electricity markets worldwide has fundamentally repositioned storage hydropower producers, necessitating a shift in their operational objective from optimizing regional energy output to maximizing revenue through active market participation [9]. Hydropower remains a critical energy source, constituting approximately 15% of global electricity generation [9]. Critically, due to their inherent operational flexibility, storage hydropower systems are indispensable assets for maintaining energy balance and stability within national power grids, especially when facilitating the integration of intermittent Renewable Energy Sources (RES) like solar and wind power [9].

The transition to a market-driven operational objective makes accurate short-term generation forecasting indispensable for two primary reasons: revenue maximization and grid stability. It enables producers to align water discharge schedules with high expected electricity price periods, optimizing revenue through a complex, non-linear process that is constrained by persistent hydrological uncertainties and the uncertain dynamics of the competitive electricity market [9]. At the same time, reliable forecasts are crucial for informed bidding in the day-ahead and intra-day markets, ensuring the timely commitment of flexible hydropower assets to support grid stability and maintain energy balance during periods of high demand or low intermittent RES output.

Cascaded hydropower systems significantly intensify the forecasting challenge. Unlike a single-plant system, a cascaded arrangement consists of multiple power plants situated sequentially along a river [9]. The power generation and water release decisions of an upstream plant directly influence the water inflow—and thus the potential generation—of all subsequent downstream plants. This creates a highly complex, coupled, non-linear system where the optimal operational schedule must account for the interdependencies and constraints of the entire chain, rather than just isolated units. Accurate short-term forecasting must therefore model the dynamic hydraulic and economic relationships across the cascade.

Given the complexities and non-linear characteristics inherent in short-term generation forecasting, particularly in coupled hydro-econometric and cascaded environments, traditional statistical methods often lack the robust modeling capabilities required to maintain high prediction precision [8].

As a result, there is substantial and growing research interest in the adoption of Artificial Intelligence (AI) and Machine Learning (ML) methodologies. AI models excel at identifying non-obvious correlations and complex, non-linear patterns within large, noisy datasets, frequently demonstrating superior accuracy over conventional models [10], [11]. Furthermore, the development of sophisticated hybrid forecasting models, which blend physics-based forecasts (such as those from numerical weather prediction) with data-driven AI/ML systems, is increasingly recognized as a vital approach for enhancing prediction skill and minimizing inherent biases [12].

The Radial Basis Function Neural Network (RBFNN) is a highly suitable Artificial Neural Network (ANN) architecture for this domain. It is distinguished by its capacity for effective generalization and simplicity in approximating nonlinear system behavior, even with a single hidden layer [16], [17]. RBFNNs are efficiently utilized in hybrid schemes for tasks like input data clustering and localized regression, and their performance in short-term forecasting can be

further optimized by combining them with meta-heuristic algorithms [16], [17].

B. Physical and Operational Characteristics of Cascaded Hydroelectric Power Plants

The foundation of hydropower optimization lies in understanding the conversion process: hydroelectric plants convert the kinetic and potential energy of flowing water into electrical energy [3]. Optimal power dispatch requires efficiently leveraging the available hydraulic head and river flows to maximize the resultant power, which is primarily determined by the flow of water through the turbine and the effective height of the waterfall [13]. Accurate short-term generation forecasting must therefore account for the gross head available, efficiency losses in the turbine-generator set, and hydraulic losses, often by integrating hydrological and climatic data to predict future water availability [3].

The complexity of management is significantly increased in a cascaded hydro system, which features multiple power plants situated sequentially along the same river [20]. This setup creates a critical spatial coupling because the outflow (released water) from any upstream plant inherently becomes the inflow for its downstream counterpart [21]. This interdependency means that upstream operational decisions—particularly release schedules—exhibit a progressive dominance over the entire chain, creating a highly non-linear system where localized downstream rules have limited influence [20]. Consequently, the core management challenge is the coordination of these interconnected facilities, which is governed by the mass balance principle (inflow minus outflow equals change in volume) and various operational thresholds.

Effective short-term forecasting and operation rely on monitoring and controlling several critical, coupled variables:

- **Reservoir Volume and Level:** Maintained within predefined minimum and maximum allowable storage levels; this level determines the available hydraulic head [14].
- **Inflow and Outflow:** The total outflow discharge comprises the crucial turbined flow (which generates power) and any intentional spillage [13].
- **Turbine Discharge and Gate Opening:** The hourly decision variables that directly control power production and, in the case of flood management, dictate spillway settings [14].

These systems face practical coordination deficits, including managing uncertainty in future inflows, overcoming structural model deficits that fail to adequately represent inter-plant coordination, and mitigating the risk of strategic market behavior by upstream firms that might restrict water flow to limit downstream rivals [20], [21].

C. Hydropower Generation Forecasting and Time Horizons

The shift toward sustainable energy necessitates the accurate prediction of electricity generation across all resource types. Hydropower Generation Forecasting (HGF) involves estimating the electrical output of hydroelectric facilities over future time intervals, a process crucial for maintaining the operational reliability and economic viability of modern power systems [26]. While HGF involves fewer direct meteorological uncertainties compared to highly intermittent sources like wind and solar, accurate short-term prediction remains critical for complex systems such as cascaded power plants to optimize water resource management and system integration. Forecasting techniques, whether applied to electricity load or generation, are conventionally stratified into distinct categories based on the temporal length of the predictive horizon, as

these intervals correspond to specific planning and operational requirements [22]. Long-Term Forecasting (LTLF) typically spans five years up to decades ahead, guiding strategic policy and power system expansion. Medium-Term Forecasting (MTLF) covers months to years ahead, supporting financial applications like energy trading and unit maintenance scheduling [23], [40]. The most operationally relevant horizon is Short-Term Forecasting (STLF), characterized by timeframes ranging from a few hours up to one or two days ahead [23]. This category is highly valuable for immediate system operation planning and coordination [40].

The core focus of short-term generation forecasting in hydropower is essential for operational security, load-sharing decisions, and maximizing economic returns in the electricity market [23]. Accurate STLF is vital for effective scheduling and dispatch, allowing operators to optimize system operations, minimize production costs, and avoid potential system overloads [26]. Furthermore, STLF supports the integration of intermittent resources by providing the necessary predictive input for hydropower to maintain grid stability and compensate for fluctuations in wind and solar output. HGF is complicated because it involves many nonlinear physical and operational variables. These include hydrological inputs like rainfall and river inflow, reservoir metrics like forebay elevation (water level), and operational controls like turbine discharge/outflow and gate openings [3], [13].

While hydropower is a conventional dispatchable source, the STLF challenge is significantly different from wind or solar, which struggle primarily with meteorological intermittency. For a cascaded hydroelectric system, the difficulties are rooted in hydraulic and spatial coupling. Specifically, the model must account for water routing time delays, which represent the highly variable and non-linear time it takes for released water to travel from an upstream dam to the reservoir of a downstream dam. It must also capture cascade effects, the precise interdependency where the outflow of one plant immediately and consequentially impacts the inflow and available head of the next plant in the chain [15].

D. Conventional Approaches and the Necessity of AI

Accurate forecasting of hydropower generation (HPG) is a fundamental prerequisite for the efficient management of water resources, capacity planning, and successful participation in dynamic energy markets [41]. Historically, hydropower and related hydrological variables were forecasted using conventional statistical and physical models [24], [41].

1) Traditional Forecasting Methods and Their Limitations:

Among the classical techniques, the AutoRegressive Integrated Moving Average (ARIMA) model is a complex linear framework widely employed in time series analysis [41], [42]. Its seasonal counterpart, the Seasonal ARIMA (SARIMA) model, is applied when the time series exhibits clear periodic or seasonal patterns [41]. Additionally, Multi-Linear Regression (MLR) and traditional hydrological/physical models represented the standard methodology used to forecast key variables like reservoir water levels [24].

Despite their mathematical simplicity and long-standing flexibility, these conventional methods face considerable limitations when applied to modern, complex hydroelectric systems [39], [41]. A core difficulty is the restricted capability of linear models (such as MLR and ARIMA) to adequately model the significant non-linearity and non-stationarity found in hydrological and environmental time series [39]. Complex, non-linear functions govern real-world systems, especially those related to coupled energy and water dynamics [39].

Traditional techniques primarily focus on forecasting linear and static processes and struggle to identify these complex non-linear structures. Consequently, they exhibit poor performance, difficulty in accommodating multiple exogenous variables, and often fail to accurately generalize patterns, resulting in substantially higher prediction errors [4], [41].

2) *Application of AI in Energy Forecasting:* The complexity arising from uncertain, dynamic, non-linear, and non-parametric variables in hydropower optimization necessitated a strategic shift toward AI and ML techniques [3], [39]. ML models demonstrate preferable adaptability in prediction, circumventing the need for pre-defined classical mathematical definitions and efficiently modeling nonlinear processes [24]. AI algorithms are now extensively applied across hydropower optimization, ranging from river flow forecasting to reservoir operation [24].

Key AI and ML models applied in energy and hydropower forecasting include:

- **Artificial Neural Networks (ANNs):** The most widely utilized ML technique, deployed to predict river flow, reservoir inflow, and short- to medium-term production due to their robustness and superior capability to derive complex non-linear relationships [3], [41]. The Radial Basis Function Neural Network (RBFNN) is a specific ANN architecture frequently investigated in forecasting literature and is the focus of this study [41].
- **Support Vector Machines (SVMs) and Regression (SVR):** Widely applied due to their demonstrated efficacy in modeling non-linear processes and their superior performance over simpler linear methods for river discharge forecasting [41].
- **Tree-based Models:** Algorithms like Random Forest (RF) and Boosted Decision Tree Regression (BDTR) constitute a significant category, successfully applied to forecasting and noted for their ability to handle smaller datasets and multiple exogenous inputs [24].

3) *Performance Superiority of AI Models:* AI models consistently demonstrate performance advantages over conventional statistical techniques in HPG time series forecasting [39]. This superiority stems from several key capabilities:

- **Modeling Non-linearity and Complex Dynamics:** AI models, particularly ANNs and SVR, display superior capability in modeling the inherently non-linear processes of HPG data, in sharp contrast to the struggles faced by simple statistical techniques like ARIMA when confronted with complex structures [39], [41].
- **Generalization and Accuracy:** Machine learning approaches are consistently found to be powerful tools that provide better accuracy and generalization than conventional methods [41]. Comparative studies frequently show that ANN and SVM models surpass ARIMA/ARIMAX techniques in prediction accuracy for hydroelectric production across various time horizons [43].
- **Adaptability to Complex Systems:** The AI approach is generally preferred because it offers a stronger capability to address the complexity of modeling time-varying systems, especially those with coupled dynamics like cascaded hydroelectric power plants, where the operation of one reservoir non-linearly affects the others downstream [3].

While deep learning models may sometimes require careful tuning of numerous hyperparameters, when properly optimized, AI algorithms generally achieve better prediction performance in most forecasting applications [39].

E. Artificial Neural Networks for Generation Forecasting

The increasing complexity and dynamic operation of modern power systems, particularly those incorporating renewable energy sources, have necessitated the development of highly accurate and adaptive forecasting methods [28]. ANNs have emerged as one of the most effective families of models for handling the inherent non-linearity and complexity found in load and generation time series data [28], [29].

1) *Conceptual Foundations and Suitability of ANNs:* An ANN is defined as a highly connected array of elementary processors, or neurons, loosely inspired by the structure and distributed processing of the biological human brain [28]. Fundamentally, an ANN functions as a non-parametric model designed to approximate the unknown function that generates observed data. The typical architecture includes an input layer, one or more hidden layers, and an output layer [28]. The network is trained by iteratively updating the synaptic weights—which store the acquired knowledge—through algorithms like backpropagation, minimizing a cost function based on the output error [28]. ANNs are exceptionally well-suited for short-term forecasting because they possess the capacity to approximate arbitrarily well a wide class of linear and nonlinear functions without requiring restrictive prior knowledge of the data-generating process [29], [44]. This capability is critical when facing non-stationary time series and abrupt environmental or operational changes that render traditional statistical methods ineffective [28].

2) *ANN Architectures and Applications in Energy Forecasting:* The ANN family includes various architectures broadly divided into Feedforward Neural Networks (FFNNs), where signals travel linearly (e.g., Multilayer Perceptron (MLP) and Radial Basis Function (RBF), and Recurrent Neural Networks (RNNs), which incorporate internal feedback loops to retain an internal state and model temporal dependencies (e.g., Long Short-Term Memory (LSTM) networks) [44].

ANNs have been extensively employed across power system operations:

- **Load and Renewable Forecasting:** ANNs are the most prominent non-parametric tool for STLF [28]. MLPNNs have also been used successfully for solar irradiance and rainfall forecasting by modeling complex non-linear meteorological variables [30], [45].
- **Hydropower Prediction:** ANNs have been successfully applied in studies to predict hydroelectric production and reservoir inflow [43], [46]. Hybrid models combining ANNs with optimization algorithms show superior accuracy in forecasting available energy [46].

3) *Advantages and Typical Limitations of ANNs:* ANNs offer significant advantages over conventional statistical techniques:

- **Non-linear Modeling:** Their primary strength is the ability to accurately approximate and model complex non-linear behaviors inherent in fluctuating and non-stationary time series data, consistently producing more accurate results [4], [29].
- **Data-Driven Generalization:** ANNs are data-driven and do not impose restrictive assumptions, enabling them to implicitly detect complex non-linear relationships between variables [30].
- **Temporal Memory:** Recurrent architectures (RNNs, LSTM) can utilize internal feedback mechanisms to integrate historical dependencies into the current prediction, a feature crucial for time-series forecasting [44].

Despite their robustness, ANNs are associated with several limitations:

- **Data Requirements:** ANNs are “greedy” models, typically requiring a substantial volume of historical data to train effectively and generalize reliably [29].
- **Overfitting and Complexity:** Improper selection of network parameters can lead to overfitting, severely compromising the ability to generalize to unseen data [?]. Extensive trial-and-error is often required to find the optimal design, such as the number of hidden neurons [45].
- **Training and Computational Cost:** Models with more than one hidden layer or complicated recurrent structures (like LSTMs) are naturally more complicated, take longer to train, and need more computing power [4].

Hybrid modeling techniques and optimization algorithms often mitigate these limitations [46]. The suitability of the specific architecture, such as the Radial Basis Function Network, must be evaluated based on the complexity and volume of the available cascaded hydropower data.

F. The Radial Basis Function Neural Network (RBFNN)

The accurate short-term generation forecasting of cascaded hydropower plants requires predictive models capable of handling complex, nonlinear system dynamics and rapidly processing large volumes of time-series data [12]. Within the general ANN framework, the RBFNN architecture provides distinct, specialized advantages for high-precision function approximation tasks such as forecasting.

1) *RBFNN Architecture and Differentiation from MLP:* The RBFNN is a particular type of feed-forward network recognized as a universal function approximator, capable of approximating complex mathematical functions to any arbitrary degree of precision [19]. This capability is shared with the widely used Multi-Layer Perceptron (MLP) network. However, the RBFNN architecture fundamentally differs from conventional MLP networks in its structure and core function [19].

The RBFNN relies on a simple three-layer structure: an input layer, a single hidden layer using non-linear radial basis functions (RBFs) as activation functions, and an output layer [19]. In contrast, the MLP typically requires multiple hidden layers to achieve comparable high-precision function approximation [19]. The core functional difference lies in the localized response of the hidden layer: each hidden neuron calculates a Gaussian function based on the Euclidean distance between the input vector and the neuron’s stored “center” [12]. This means RBFs function as local receptors, providing a strong output only when the input value is close to the node’s center, differentiating it from the global response characteristic of MLP neurons [19]. [19].

2) *Fundamentals of RBFNN Parameter Determination:* The overall training task of an RBFNN is typically implemented through two-phase learning [18].

- **RBF Layer Parameters (Centers and Widths):** The first phase involves determining the centers and widths or spreads of the RBF layer, often implemented through unsupervised learning [48].
 - **Centers:** These are vectors representing prototypes of the input data, and their choice is critical [19]. Common methods for setting centers include random sampling or k-means clustering, which places centers in areas that best represent the overall data distribution [19].

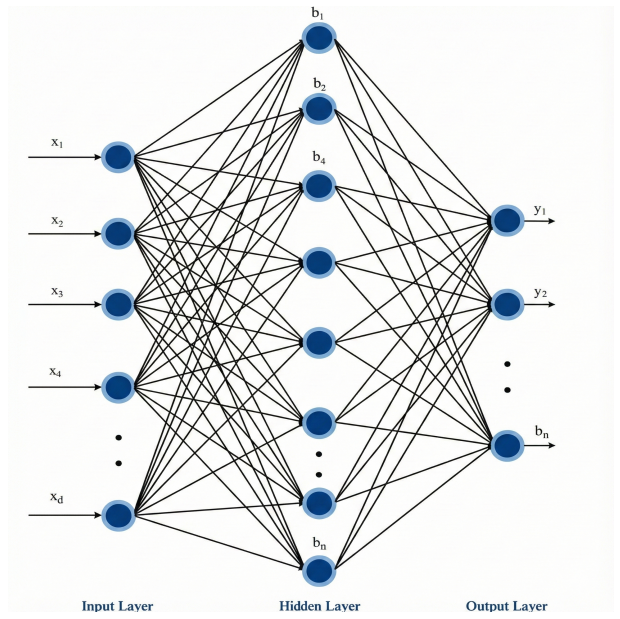


Fig. 2: Architecture of the RBFNN [47].

- **Widths (Spreads):** The width parameter controls the shape of the Gaussian basis function, determining the extent of the node’s localized influence [19].

- **Output Weights:** The second phase involves determining the weights between the RBF layer and the output layer [19]. This is the only part of the RBFNN that requires supervised learning. Optimal weights can be determined either through iterative methods like gradient descent or, critically, by direct calculation using a pseudoinverse approach, which efficiently finds a minimum-norm weight solution for the given data [19].

3) *RBFNN Success in Forecasting and Function Approximation:* RBFNNs have proven versatile and powerful across a broad array of practical applications, confirming their utility in high-precision tasks [19]. Successes span numerous domains, including the solution of partial differential equations, computer vision, and medical classification [48].

For predicting dynamic systems, RBFNNs demonstrate strong performance in various time-series analysis and forecasting contexts:

- **Load Forecasting:** RBFNNs have been successfully implemented for short-term load forecasting in electric power systems, demonstrating high accuracy in predicting loads 10 minutes ahead, an application that inherently deals with non-linearity and periodicity [12].
- **System Prediction:** RBFNNs have been utilized for forecasting power in wind energy conversion systems and for categorizing high-resolution electrocardiograms (ECGs), demonstrating their effectiveness in processing 1D time-series data [12], [18].
- **Complex Function Approximation:** RBFNNs have undergone extensive testing on intricate non-linear mathematical optimization functions and real-world datasets, consistently producing precise predictions [19].

4) *Suitability for Short-Term Hydro Generation Forecasting:* The intrinsic characteristics of RBFNNs make them particularly well-suited for the specific demands of short-term generation forecasting in cascaded hydropower systems:

- **Fast and Efficient Training:** The rapid training process of RBF networks—often orders of magnitude faster than MLPs—is a critical asset for short-term forecasting models that require frequent retraining or rapid iterative adjustment to reflect current operational conditions [12], [19]. Furthermore, RBF networks typically avoid the issue of being trapped in local minima often encountered in backpropagation-trained ANNs [12].
- **High Capacity for Non-Linear Dynamics:** As universal function approximators, RBF networks excel at accurately approximating the highly complex, non-linear dynamics characteristic of cascaded hydro systems, which are governed by reservoir interactions and hydraulic constraints [19]. This allows for crucial non-linear mapping between input stimuli and power output.
- **Local Approximation and Interpretability:** The localized nature of the RBF functions allows individual nodes to specialize in approximating the function within a specific region of the input space [19]. This local tuning helps the network effectively handle noisy or sparse data points common in real-world operational environments. Additionally, the RBF centers can often be interpreted as representative patterns of the data set, providing a degree of interpretability that is valuable in technical systems [18].

The combination of computational speed, powerful non-linear modeling capability, and localized learning makes the RBFNN architecture a robust and advantageous choice for the specific demands of short-term hydroelectric generation forecasting, prioritizing efficiency and accuracy in handling complex operational data.

5) RBFNN Training and Hyperparameter Tuning Methodology:

The successful application of the RBFNN for short-term generation forecasting requires a rigorous training protocol focusing on effective parameter optimization and systematic hyperparameter tuning. The RBFNN's efficiency stems from its ability to employ hybrid learning where the hidden node parameters are set prior to optimizing the output weights [49].

A. Training RBFNN Parameters The RBFNN training aims to find the optimal configuration of the centers, spreads, and output weights that minimize the Mean Square Error (MSE) [50].

- **Centers Selection and Clustering:** The K-means (KM) algorithm is widely adopted for unsupervised center selection due to its speed and simplicity, grouping input cases into clusters where the centroids become the RBF centers [49]. However, standard K-means is susceptible to the initial selection of centers and may not always extract the most accurate prototypes from complex time-series datasets, prompting the investigation of improved methods, such as the Distance-Weighted K-means (DWKM) algorithm [49].
- **Traditional and Gradient-Based Optimization:** The Gradient Descent (GD) algorithm is a conventional derivative-based method often used for RBFNN training [50]. However, like many derivative-based methods, GD frequently gets trapped in local minima and is generally outperformed by more advanced search techniques [51].
- **Metaheuristic and Hybrid Optimization:** To achieve global optimization and efficiently determine all complex parameters simultaneously, metaheuristic and swarm intelligence algorithms have become powerful alternatives [50], [51]. Algorithms like the Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and the Firefly Algorithm (FFA) model the entire set of RBFNN parameters (weights, spreads, centers) as a

single solution vector. These techniques utilize randomized search to efficiently explore the solution space, demonstrating competitive or superior performance to traditional GD and GA algorithms when training RBFNNs for time-series prediction [51](Hamadneh, 2020).

B. Hyperparameter Tuning Strategies

Hyperparameter tuning focuses on selecting values for parameters that control the learning process, such as the number of hidden neurons (which equals the number of centers) and the spread parameter (σ) [52].

- **Tuning via Experimentation and Validation:** The most practical and common strategy involves extensive trial-and-error experimentation combined with the use of a validation dataset [50]. Researchers typically test RBFNNs with varying numbers of hidden neurons to identify the configuration that yields the optimal performance metric on the validation set [50]. This approach ensures the hyperparameter selection is robust and prevents overfitting to the training data.
- **Tuning via Optimization Algorithms and Criteria:** In implementations utilizing metaheuristic training (PSO, FFA), the hyperparameter selection is often intrinsically part of the optimization process, as the algorithm directly searches for the best combination of centers and spreads alongside the weights [50]. To formally evaluate the best resulting architecture, model selection criteria like the Schwarz Bayesian criterion (SBC) are utilized to assess the trade-off between minimal error and network complexity [51]. Cross-validation methods may also be used to ensure the chosen architecture is not biased towards a particular subset of the data [50].

6) Dataset Splitting in Time-Series Forecasting: In studies that use time-series forecasting, the dataset must be split up in strict chronological order so that the model only learns from the past to make predictions about the future. Unlike general machine learning problems, random splitting is inappropriate, as it introduces future information into the training phase, leading to optimistically biased performance estimates.

The dataset is segmented into three distinct, non-overlapping subsets based on time:

- **Training Set (70%):** Comprises the earliest data and is used for fitting the model parameters (centers, spreads, and weights).
- **Validation Set (10–15%):** Follows the training set chronologically and is used to tune the model hyperparameters (selecting the optimal number of hidden neurons or spread value).
- **Testing Set (10–15%):** Comprises the latest data, reserved exclusively to provide a final, unbiased evaluation of the model's generalization capability on entirely unseen, recent conditions [52].

In quantitative studies, common ratios like 70% for training and 30% for testing (or 70–15–15% when a separate validation set is used) are frequently employed, provided this splitting is done sequentially [51]. This rigorous partitioning methodology is essential for developing a reliable short-term generation forecasting model for cascaded hydroelectric power plants.

G. Data Preprocessing and Quality Control

The performance, stability, and reliability of AI models, especially the RBFNN, are fundamentally contingent upon the volume, quality, and diversity of the input data [34]. Raw data collected from real-world applications are susceptible to numerous issues, necessitating

robust preprocessing before being utilized for short-term generation forecasting [34].

1) *Data Problems in Hydropower Time-Series*: Hydropower generation and associated meteorological and operational observations manifest as complex multivariate time-series data [6]. These datasets frequently exhibit intrinsic issues that complicate the forecasting process:

- **Missing Values and Irregular Sampling**: Missing data are a common problem resulting from technical difficulties, sensor failures, or network disconnections [32]. These failures can create large gaps, sometimes stretching for weeks [31]. Additionally, data acquisition may suffer from irregular sampling or asymmetric measurements due to transmission delays, complicating the time-alignment of observations [6].
- **Sensor Noise and Inconsistency**: Raw data can be affected by noise and corruption caused by how sensors work or how they communicate [31]. Data inconsistency occurs when information gathered from various heterogeneous sources differs in standards or detail [34]. For sensitive machine learning models like ANNs, the presence of noisy and unreliable data significantly impedes the knowledge discovery process during training [33].

2) *Strategies for Handling Missing Values and Imputation*:

Addressing missing data is critical because models that rely on continuous time series, such as RBFNNs, require input vectors that are complete and representative [31].

- **Simple Univariate Interpolation**: For filling small data gaps, Linear Interpolation (LIN) is highly effective, assuming a linear function between the nearest good measurements [31]. Spline Interpolation (SPL), such as using cubic splines, expands on this by creating a smooth, piecewise polynomial function that is less susceptible to high-variance periods [32]. Other basic methods include replacement using the mean, median, or mode [34].
- **Advanced Model-Based and Multivariate Imputation**: For handling larger gaps or complex patterns, sophisticated methods are required. Multivariate methods often yield superior results for large gaps by leveraging inherent periodicity and relationships within the dataset [31]. These include:
 - *K-Nearest Neighbors (KNN)*: Imputes values using a weighted mean of the k most similar data rows, proving accurate for medium gaps [31].
 - *Matrix Factorization (MF)*: Decomposes the time series matrix to infer missing values, offering the most accurate imputations for the largest gaps across different sensor categories and missing rates [31].
 - *Autoregressive (AR) Models*: Calculate missing values as forecasts, although this method requires uninterrupted segments of data for training coefficients [32].
- **Moving Average and Sliding Window**: The Weighted Moving Average (WMA) is widely adopted for smoothing highly non-linear time series data, where the highest weight is assigned to the most recent data to enhance trend detection [53]. This is often combined with a sliding window concept, where a fixed-size window aggregates historical data points to generate new, smoother input observations for the RBFNN [53].
- **General Filtering**: Other methods include binning, which reduces the impact of minor observation errors by replacing binned values with derived general values [34]. More generally, sophisticated algorithms can be used that explicitly filter,

modify, or relabel noisy data while preserving essential data features [34].

3) *The Crucial Role of Robust Data Preprocessing*: Noise Reduction and Data Smoothing Techniques Minimizing the presence of noise is essential prior to training sensitive AI models such as RBFNNs, as noise significantly impedes the knowledge discovery process [34]. Filtering and smoothing techniques improve trend detection and generalization performance by highlighting essential features [53].

- **Moving Average and Sliding Window**: The Weighted Moving Average (WMA) is widely adopted for smoothing highly non-linear time series data, where the highest weight is assigned to the most recent data to enhance trend detection [53]. This is often combined with a sliding window concept, where a fixed-size window aggregates historical data points to generate new, smoother input observations for the RBFNN [53].
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3) *The Crucial Role of Robust Data Preprocessing*: Robust preprocessing is not merely a preliminary step but a critical determinant of the eventual success, stability, and reliability of AI-based forecasting models [33]. Poor data quality significantly affects the accuracy and reliability of predictions, potentially leading to false conclusions [34].

The substantial time dedicated to initial data preparation (estimated at 50% to 80% of the entire process) underscores its importance [34]. Data cleaning ensures the resultant dataset meets critical quality criteria, including validity, accuracy, completeness, consistency, and uniformity [34]. Ultimately, ensuring that the training data is clean and representative maximizes the RBFNN's ability to extract meaningful non-linear information, thereby guaranteeing stable and reliable AI-based forecasts necessary for cascaded hydropower operation.

H. Evaluation Metrics

The validation of forecasting models requires robust statistical metrics to assess prediction performance. This study employs standard metrics commonly used across hydrological and energy forecasting literature: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2) [35]–[37].

- **Root Mean Squared Error** is derived from the Mean Squared Error (MSE) and uses the L_2 -norm, involving the squaring of prediction errors [35], [37]. Because errors are squared, RMSE is inherently more sensitive to outliers, giving larger weights to points with high deviations [35]. Minimizing RMSE is statistically optimal if errors follow a normal (Gaussian) distribution [37]. RMSE yields a metric with the same units as the target variable [37].
- **Mean Absolute Error** minimizes the L_1 -norm, which uses a linear treatment of errors [Hodson, 2022]. MAE is recognized as a robust measure because the L_1 -norm tends to smooth out the penalty applied to extreme outliers, making it less sensitive to severe forecasting mistakes compared to RMSE [35], [37]. MAE is mathematically optimal when errors follow a Laplacian distribution [37].

- Mean Absolute Percentage Error is widely utilized and provides a highly intuitive interpretation in terms of relative error [35], [36]. This metric is particularly useful when the application emphasizes sensitivity to proportional variations [36]. A critical consideration is that MAPE is restricted to cases where the target variable is strictly positive and is known to be heavily biased towards low forecasts [35].
- Coefficient of Determination quantifies the proportion of variance in the dependent variable that is predictable from the independent variables [35]. R^2 offers a bounded, scale-independent assessment of fit [35], ranging from negative infinity up to 1 (perfect prediction). A value like $R^2 = 0.8$ clearly suggests a very good regression model performance, irrespective of the predicted generation output's scale [35].

These metrics are essential for a comprehensive evaluation, where MAPE is critical for assessing relative accuracy (desirable in generation forecasting) and R^2 provides an absolute measure of goodness-of-fit for comparing the RBFNN against alternative methods [35], [36].

I. Research Gaps

Despite the proven success of AI in general forecasting, several specific methodological and domain-focused gaps remain when approaching AI-based short-term generation forecasting for cascaded hydroelectric power plants. These gaps define the need for the present thesis:

- 1) **Limited Focus on Cascaded Systems as a Forecasting Problem**
The existing literature often treats cascaded hydropower as an optimization problem where forecasts are merely external inputs [13], [15]. Conversely, most hydropower forecasting studies focus on single-reservoir or individual station output, neglecting the core challenge of explicitly modeling the hydraulic coupling, routing delays, and head dependencies simultaneously across an entire cascade [3], [4]. This leaves a gap in dedicated data-driven models for concurrent, coupled generation forecasting in cascaded systems.
- 2) **Lack of Focused One-Day-Ahead Operational Resolution**
While short-term horizons are recognized as critical, most hydropower forecasting is geared toward daily or longer resolutions for medium-term planning [24], [25]. There is a clear lack of research explicitly targeting the one-day-ahead (24-hour) horizon with sufficient temporal granularity. This specific resolution is essential for crucial tasks like market bidding, spill management, and real-time coordination of reservoir releases, a gap highlighted by comparisons with advanced load and wind/solar forecasting [26], [27].
- 3) **Underutilization of RBFNNs**
Despite their documented strengths as universal approximators with fast training speed, localized basis functions, and robustness in non-linear regression [18], [19], RBFNNs are underutilized as the primary architecture for hydropower generation forecasting. Most studies favor MLPs, SVMs, or tree-based models [41], [46]. This presents a gap in systematically applying and comparing the RBFNN architecture for one-day-ahead generation forecasting in complex cascaded systems.
- 4) **Simple Feature Integration Lacking Cascade Physics**
Many existing AI models for hydropower rely on relatively simple input sets. This often fails to fully exploit multivariate operational features that encode the physical coupling central to cascaded operation. Specifically, there is a gap in models that

comprehensively integrate upstream releases, inter-reservoir travel time, and dynamic reservoir head across the cascade, which are critical physical drivers of short-term generation [13], [14].

5) Preprocessing and Evaluation

A methodological gap exists concerning the documentation and consistent application of complete data quality pipelines. While the need for handling missing values, sensor noise, and chronological data splitting is stressed [31], [52], few studies provide a unified, reproducible preprocessing protocol tailored for multivariate cascaded hydropower data—combining advanced imputation, smoothing, sliding-window feature construction, and strictly chronological splitting for reliable time-series evaluation.

In response to these gaps, the present study develops an AI-based short-term (one-day-ahead) generation forecasting framework for a cascaded hydropower system using a Radial Basis Function Neural Network (RBFNN). The proposed model (i) treats the cascade as a coupled forecasting problem, (ii) explicitly incorporates cascade-relevant physical features such as upstream releases, travel time, and reservoir head, and (iii) employs a rigorous, time-consistent data preprocessing and evaluation protocol using MAPE. This framework is intended to serve as a practical and methodologically transparent benchmark for operational forecasting in cascaded hydroelectric power plants.

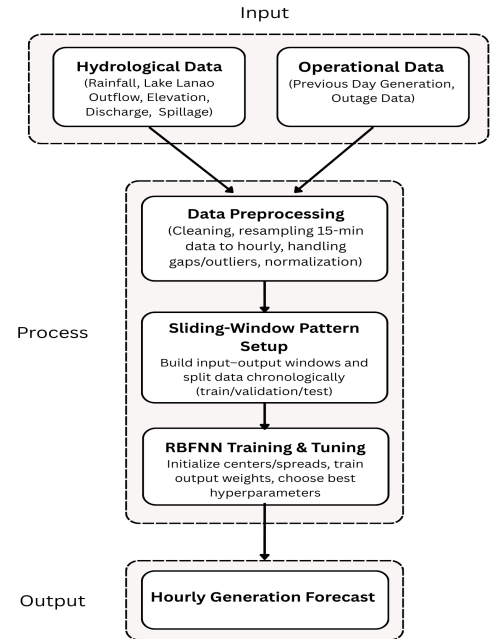


Fig. 3: Conceptual Framework For Hourly Generation Forecast

III. METHODOLOGY

A. Introduction

This chapter will describe the methodological framework for the study, “AI-Based Short-Term Generation Forecasting of Cascaded Hydroelectric Power Plants.” The study will employ a developmental–experimental design in which historical operation data of the Agus hydropower cascade will be used to train and

evaluate a Radial Basis Function Neural Network (RBFNN) for day-ahead generation forecasting.

The research procedure will follow the workflow shown in Figure 1, consisting of four main stages: data collection and preparation, RBFNN model development and training, day-ahead forecasted generation, and forecast evaluation and benchmark comparison.

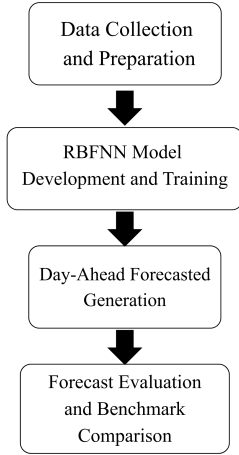


Fig. 4: Workflow of the Study.

B. Research Design

This study will adopt a simulation-based, data-driven forecasting design using a supervised learning framework. The main objective will be to learn a nonlinear mapping between historical hydropower operating variables and next-day hourly generation for each plant and the total generation in the Agus 1–7 cascade using a Radial Basis Function Neural Network (RBFNN).

RBFNNs will be used because they have demonstrated strong performance in nonlinear time-series forecasting across power and water resources applications, including short-term PV power forecasting [54], [55], short-term load forecasting [47], and rainfall prediction [56].

C. Materials and Apparatus

1) *Software Environment*: The following software tools will be used:

- Microsoft Excel will be utilized to collate and organize the raw historical operational data (generation, elevation, spillage and gate openings) obtained from the plant operator.
- Python will be the primary programming language used to develop, train, and evaluate the RBFNN forecasting model.
- Google Colab will function as the cloud-based Integrated Development Environment (IDE) for writing and executing code.

D. Data Collection and Preparation

1) *Study Area and Target Variable*: The study focuses specifically on the Agus 1–7 hydroelectric power plants and the operational dynamics of their associated reservoirs and river reaches. The primary target variable for the short-term forecasting model is the hourly active power generation (P_t) measured in megawatts (MW). This generation will be predicted for each individual plant, as well as for the crucial total cascaded output. The model's objective is to provide a consistent 24-hour-ahead forecast horizon, predicting the hourly output for the entire following day.

2) *Data Sources and Collected Variables*: Historical data have been requested from the plant operator for a multi-year period. For each plant, the following variables will be collected:

- Fifteen-minute elevation data,
- Fifteen-minute plant generation data,
- Spillway gate opening data,
- Governor gate opening data,
- Riparian flow data,
- Inflow and outflow data,
- Other relevant operational and hydrological data.

All required historical operational datasets for the Agus 1–7 cascaded power plants have been successfully collected and are fully available. These data were initially recorded at a high-resolution 15-minute interval and stored in a unified database, indexed by plant, date, and time. This high-resolution data will be systematically resampled to an hourly time base by calculating averages, ensuring the forecasting model operates on the required hourly time step for the day-ahead forecast while simultaneously preserving the underlying operational dynamics captured by the high-resolution measurements.

3) *Data Cleaning and Handling Missing Values*: The following steps will be applied:

- Consistency checks. Basic rules (non-negative generation, reservoir level within physical limits, discharge continuity between cascade plants) will be used to flag abnormal or inconsistent readings, which will either be corrected (if obvious logging errors) or treated as missing.
- Gap filling. Short gaps will be filled using linear interpolation replacement within the same day type. Longer gaps will be excluded from training sequences to avoid introducing artificial patterns.
- Outlier treatment. Extreme values outside a predefined percentile range (below the 1st or above the 99th percentile) will be examined. If they correspond to rare but real events, they will be retained; otherwise, they will be either winsorized or removed.

4) *Normalization*: To improve numerical stability and to match typical RBFNN practices in hydrological and energy forecasting, all continuous variables will be normalized to $[0, 1]$, a dimensionless scale, before being fed into the RBFNN. A min–max normalization will be applied:

$$x_{\text{nom}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where x is the original value, $x_{\min}$ and $x_{\max}$ are the minimum and maximum observed values for that variable, respectively, and x_{nom} is the normalized value [47], [55].

5) *Input–Output Pattern Construction*: The forecasting problem will be formulated in sliding-window form, where the inputs correspond to recent historical states and the target output is the future generation to be predicted. Inspired by sliding-window RBFNN time-series prediction methods [47], [53], each input vector will be built as:

$$\mathbf{x}_i = [P_{i-1}, P_{i-2}, \dots, P_{i-L_P}, H_{i-1}, \dots, H_{i-L_H}, Q_{i-1}, \dots, Q_{i-L_Q}, \dots] \quad (2)$$

while the target will be the plant generation $P_{t+\Delta}$, where Δ corresponds to one hour up to 24 hours ahead, depending on the forecast horizon design. The lag lengths L_P , L_H , and L_Q will be selected based on autocorrelation analysis and prior findings on

effective lag structures for short-term load and rainfall forecasting [47], [56].

6) *Dataset Partitioning*: To prevent information leakage and to mimic real-world deployment, the dataset will be split chronologically into three subsets:

- Training set—approximately 70% of the earliest data, used to estimate the RBFNN parameters;
- Validation set—approximately 15% of the subsequent data, used for hyperparameter tuning and early stopping;
- Testing set—approximately 15% of the most recent data, reserved for final unbiased evaluation of the forecasting performance.

This chronological splitting strategy will follow practices used in time-series forecasting studies in energy systems [47], [55].

E. RBFNN Model Development and Training

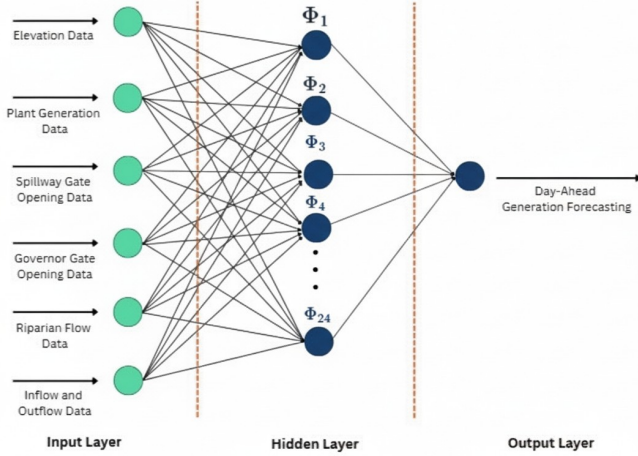


Fig. 5: Architecture of the RBF neural network based generation forecasting method [47].

1) *Network Architecture*: The proposed forecasting model will be based on a three-layer radial basis function neural network:

a) *Input layer*.: Each neuron will correspond to an input feature (past generation, reservoir level, gate opening, spillway, riparian flow). The dimension m of the input vector will equal the total number of features in the chosen window [47], [55].

b) *Hidden layer*.: The hidden layer will consist of q radial basis neurons. Each hidden neuron j will compute a Gaussian radial basis function of the form

$$\phi_j(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}_j\|^2}{2\sigma_j^2}\right) \quad (3)$$

where $\boldsymbol{\mu}_j$ is the center and σ_j is the width (spread) of the j -th basis function [47], [55].

c) *Output layer*.: The output layer will perform a linear combination of hidden activations:

$$y_k(\mathbf{x}) = \sum_{j=1}^q w_{kj} \phi_j(\mathbf{x}) + \theta_k \quad (4)$$

where y_k is the k -th output (specific plant's generation at a specific hour), w_{kj} are the connection weights, and θ_k is a bias term [47].

The number of hidden neurons q and the architecture (single vs. multi-output) will be tuned empirically to balance model complexity and generalization performance, consistent with findings in RBFNN rainfall and PV forecasting [54]–[56]. [19].

2) *Parameter Initialization and Training*: The RBFNN training will proceed in two stages, following common practice in the literature [47], [54], [56]:

a) *Hidden-layer parameter initialization*.:

- Centers $\boldsymbol{\mu}_j$ will be initialized using a clustering algorithm applied to the training inputs.
- Spreads σ_j will be derived from average distances between centers or tuned as hyperparameters.

b) *Output-layer training*.:

- With centers and spreads fixed, the output weights w_{kj} and biases θ_k will be obtained by solving a regularized least-squares problem.
- Alternative optimization strategies, such as evolutionary or swarm-based algorithms [47], [54], may be considered if basic training does not yield satisfactory performance.

Training will minimize a loss function based on the difference between predicted and actual next-day generation.

3) *Hyperparameter Tuning and Model Selection*: The following hyperparameters will be tuned using the validation set:

- Number of hidden neurons q ;
- Spread parameter σ_j (if shared or parameterized);
- Regularization strength for output-layer training;
- Length of the input window;
- Choice of input feature subset.

For each configuration, the model will be re-trained on the training set and evaluated on the validation set. The best-performing configuration will be selected using MAPE. [55], [56].

F. Day-Ahead Forecasted Generation

Once the RBFNN has been trained and validated, it will be used to produce day-ahead forecasts for the cascaded hydropower system. For a forecast day ($D+1$), all required input variables (recent hourly generations, reservoir levels, and inflows up to the end of day D) will be prepared and normalized using the scaling parameters from the training stage. The RBFNN will then generate hourly forecasts $\hat{P}_t$ for each hour of day ($D+1$).

To enhance accuracy for non-stationary hydrologic conditions, the training patterns used for each forecast may be restricted to historical “similar days” that exhibit comparable hydrometeorological characteristics, adapting the similar-day RBFNN approach used in PV power forecasting [55].

The result of this stage will be a sequence of day-ahead forecasted generation profiles for all Agus plants and for total cascade output, which will be compared against actual measurements in the final stage.

G. Forecast Evaluation and Benchmark Comparison

The data analysis plan will specify the statistical and performance metrics that will be used to evaluate the forecasting model, borrowing from the metrics commonly used in RBFNN-based rainfall, PV, and load forecasting [47], [55], [56].

1) *Error Metrics*: The Mean Absolute Percentage Error (MAPE) will be computed for each plant and for the entire cascade:

$$\text{MAPE} = \frac{100}{T} \sum_{t=1}^T \left| \frac{P_t - \hat{P}_t}{P_t} \right|. \quad (5)$$

where P_t is actual generation, $\hat{P}_t$ is forecasted generation, $\bar{P}$ is the mean of actual generation, and T is the number of forecasted hours [55], [56].

2) *Comparative Analysis*: The RBFNN forecasts will be compared against one or more benchmark models, such as:

- Persistence model: comparing next-day generation with the actual generation;
- Simple statistical model: comparing against a linear regression model.

The comparison will help demonstrate whether the AI-based approach offers a meaningful improvement over simple baselines, as is typically done in neural forecasting studies [47], [55].

H. Data Management Plan (DMP)

The study will implement a Data Management Plan (DMP) to ensure that all datasets, scripts, and results are well organized, reproducible, and protected throughout the thesis lifecycle. The DMP will not be treated as mere paperwork; it will be used as a practical tool that guides how the Agus cascade data will be handled from initial access up to long-term archiving.

- **Organize the Data**
Specify how each type of data file will be named, stored, and versioned. Descriptive file names and a structured folder hierarchy will be adopted so that any figure or table in the study can be traced back to its exact source dataset and script. A brief README file will describe the meaning of each folder and file.
- **Ensure Reproducibility**
All preprocessing, model training, and evaluation steps will be implemented in documented scripts with clear comments and fixed random seeds where applicable. The DMP will record software versions, important hyperparameters, and the exact train-validation-test splits used in the experiments. This will provide a roadmap that will allow the adviser, examiners, or future researchers to re-run the RBFNN forecasting workflow and verify or extend the results.
- **Data Protection**
The DMP will define where files will be stored and how they will be backed up. Working copies of data and code will be kept on a password-protected computer, while regular backups will be maintained on an institutional drive or encrypted cloud storage. The plan will specify backup frequency and recovery steps in case of hardware failure or accidental deletion, ensuring that months or years of collected and processed Agus cascade data will not be lost due to a single incident. Access to full datasets will be limited to the researcher and adviser, and any data shared outside this team will be anonymized or aggregated in accordance with institutional and utility confidentiality requirements.

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